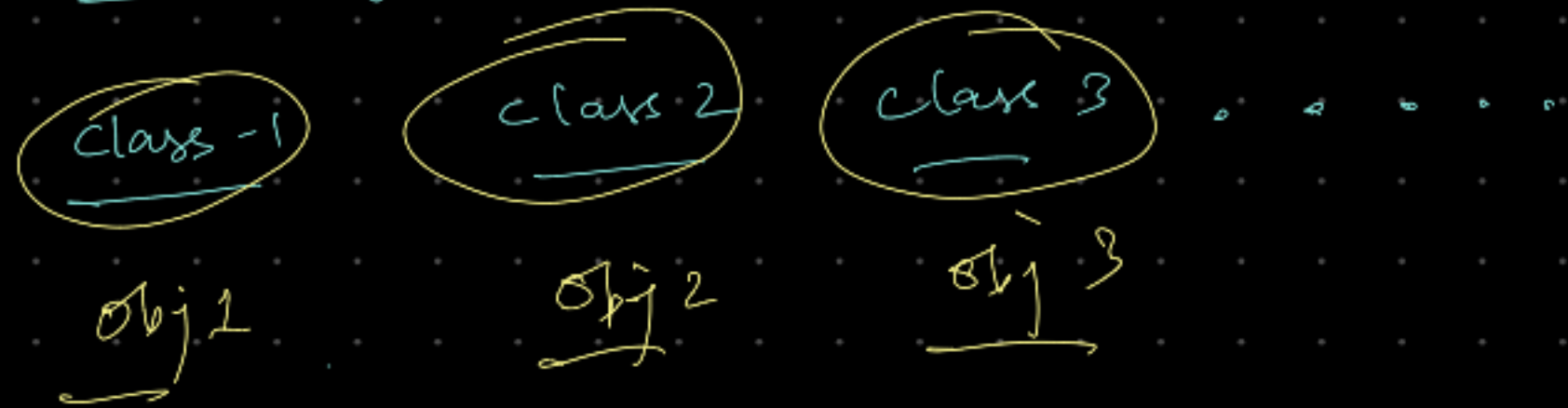


# factory Design Pattern

↳ new keyword



```
f(choice) {  
  if (choice == 1) {  
    return new C1();  
  }  
  else if (choice == 2) {  
    return new C2();  
  }  
  else {  
    return new C3();  
  }  
}
```

client  
→ factory  
 $obj_1 = f(1);$   
 $obj_2 = f(2);$   
 $obj_3 = f(3);$

## factory Pattern

Simple factory  
Not really a Design Pattern  
breaks OCP

factory Method

Abstract Method

### Simple

```
factory {  
  if  
  else  
  else
```

factory {

hashmap → choice → objects <int, obj>

1 → obj1  
2 → obj2, 3 → obj3, 4 → obj4

```
getObj(choice) {  
  map.get(choice);  
}
```

not changing

### Use-Case

Game → Levels → Enemies

Diff levels might have to create diff enemies